



Uncertainty Quantification in Climate Forecasting: How Bayesian Structural Time Series Models Excel Over Traditional and Data-Driven Techniques

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Abstract

This study investigates the efficacy of Bayesian Structural Time Series (BSTS) models for forecasting seasonal climatic variables such as rainfall, maximum temperature, and minimum temperature. The data were sourced from the World Bank Climate Change Knowledge Portal covering the period from January 1901 to December 2022, representing monthly climate records for India. Using this century-long dataset, the study compares Bayesian methods (BSTS and BSTSX) with traditional statistical methods (SARIMA, SARIMAX) and advanced machine learning approaches (XGBoost, LSTM). Lower values of root mean square error (RMSE), mean absolute error (MAE), mean absolute percentage error (MAPE), and mean absolute scaled error (MASE) were used to evaluate each model. The study reveals that Bayesian models outperform competing methods in terms of accuracy and interpretability for all three seasonal time series. The Bayesian methods improved forecast accuracy by approximately 59–66% for the rainfall series, 90–93% for maximum temperature, and 40–45% for minimum temperature, relative to competing models, based on MAPE computed from the testing set. It is acknowledged that SARIMA achieved marginally lower testing-set RMSE (0.516) and MAPE (2.230) than BSTS (RMSE: 0.614; MAPE: 2.935) for the minimum temperature series. The results show that the Bayesian structural framework is a reliable tool for climate-sensitive applications, allowing for precise forecasts that support strategies for climate adaptation, water resource management, and agricultural planning.

Keywords Rainfall · Temperature · BSTS · BSTSX · Statistical model · Machine learning

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Extended author information available on the last page of the article

1 Introduction

The climate has changed over the past few decades, mostly in terms of temperature and rainfall, which has a significant effect on environmental sustainability, water resource management, and agricultural productivity. It is essential to comprehend and predict these factors in order to effectively plan and decisions-making scenarios, as well as to reduce the hazards associated with climate oscillations (Rawat et al. 2022). Precise forecasts of rainfall and temperature can enable decision-making by stakeholders, especially in areas that rely significantly on climate-sensitive sectors like agriculture. Seasonal time series forecasting is important in this setting because it helps to predict fluctuations in weather patterns, which are significant for crop planning, disaster preparedness, and climate resilience activities. Many researchers have developed different statistical and machine learning models for forecasting the integration of climatic changes.

Despite breakthroughs in climate forecasting, considerable research gaps exist. A significant portion of the literature is based on traditional time series models such as ARIMA (Auto Regressive Integrated Moving Average) or seasonal variants such as SARIMA (Seasonal ARIMA), BATS and TBATS models, have proven effective for capturing linear trends and seasonality (Ray et al. 2021). Dabral and Murry (2017) examined the monthly, weekly and daily rainfall prediction using SARIMA model. The hydrological forecasts were also estimated using a hybrid nonlinear GARCH model (Karami et al. 2024) and β SARMA (Beta Seasonal Autoregressive Moving Average) model (Shad et al. 2023) for incorporating seasonal dynamics and non-Gaussian behaviour. The limitation of these models lies in the incomplete integration of exogenous variables, or outside influences, such as atmospheric pressure, concentrations of greenhouse gases, maximum and minimum temperatures, etc., that can affect patterns of rainfall (Shah et al. 2024). Although SARIMAX (Seasonal ARIMA with Exogenous Regressors) incorporates external elements in an attempt to close this gap, its efficacy is restricted in the case of highly complicated and non-linear connections (Mulla et al. 2024). The parametric and non-parametric models also used to estimate the prediction behaviour of hydrological information (Lama et al. 2021). Also, the hybrid approach SARIMA-GARCH model was developed using transformed hydrological data for better prediction (Pandey et al. 2019).

In addition to these conventional statistical models, a machine learning technique also used to forecast the pattern of climate change. The estimation of complex rainfall series was commonly achieved using machine learning such as non-parametric time delay neural network (TDNN) model (Benmahdjoub et al. 2013; Lama et al. 2021; Ray et al. 2023a, b). Nonlinear autoregressive neural network (NARNN) model was also employed to assess the forecasting nature of the nonlinear patterns and interactions of hydrological series (Rawat et al. 2022). The evaluation of the pattern of climate changes can also be done using the long short-term memory (LSTM) model or a distinct hybrid technique of LSTM model (Chamatidis et al. 2023; Haq et al. 2021; Zheng et al. 2024; Coutinho et al. 2025). Ouma et al. (2022) used satellite-based meteorological data from the Nzoia hydrologic basin to study rainfall and runoff trend analysis using LSTM and wavelet neural network model. The LSTM model provide more superiority than WNN as the RMSE value improved from 21 to 15 mm. A comparative analysis was done by Barrera-Animas et al. (2022) among the modern machine learning models such as XGboost, AutoML, LSTM, Stacked-LSTM,

Bidirectional-LSTM for hydrological prediction. Shrivastava et al. (2023) utilized MPR and DNN model comparing with the number of input feature for temperature prediction. Yang et al. (2017) utilized a hybrid CFCC-LSTM model, which outperformed SVM, SVR, and FC-LSTM models for sea surface temperature prediction. Wani et al. (2024) provided a multidimensional approach to estimate the complex climate change trend using deep learning. Another hybrid approach i.e. CNN-LSTM model developed by Hou et al. (2022) provided best goodness of fit for hourly air temperature prediction. A novel FORSEER methodology also utilized to get the best performance matrix in weekly temperature series (Estrada-Patiño et al. 2024). Another advanced machine learning XGBoost method was utilized to forecast complex time series meteorological data (Mishra et al. 2024; Dong et al. 2023). Liyew and Melese (2021) employed the XGBoost algorithm to forecast daily rainfall data. They discovered that XGBoost gradient descent methods outperformed RF and MLR in terms of error accuracy. Further, Kumar et al. (2025) conducted a comprehensive comparative study between machine learning models for forecasting daily and weekly rainfall of selected diverse geographical regions of India.

While extant approaches demonstrate varying degrees of forecasting efficacy, critical methodological limitations persist. Conventional time series models (e.g., SARIMAX) exhibit constrained capacity for modeling nonlinear interactions between climatic variables and their exogenous drivers, while machine learning architectures (e.g., LSTM, XGBoost) frequently lack proper uncertainty quantification mechanisms. Notably, Bayesian structural time series (BSTS) frameworks, which inherently accommodate parameter uncertainty through probabilistic formulations while maintaining interpretable decomposition of trend, seasonality, and exogenous effects, remain insufficiently explored for seasonal climate forecasting. This study addresses these methodological shortcomings through a rigorous comparative evaluation of BSTS against both classical and machine learning paradigms, incorporating dynamic exogenous variables while maintaining proper probabilistic foundations. In literature, Bayesian structural time series (BSTS) model are employed to forecast seasonal time series data associated with climatic variables like temperature and rainfall (Scott and Varian 2014). The BSTS model is a more recent development in the area, combining the advantages of classical statistical modeling with the adaptability of Bayesian inference (Das and Chanda 2020). This model are particularly well-suited for time series data because they can describe both observed and unseen components, such as trend, seasonality, and irregular fluctuations (Harrison and Stevens 1971). Song et al. (2022) investigated a two-phase Bayesian model updating and class selection approach for nonlinear structures based on NNMs that have been established through experimentation. Molina et al. (2025) designed an intelligent data-driven tool called Bayesian Stochastic Rainfall Generator (BSRG) for improving the rainfall forecast. They concluded that cluster-specific characteristics significantly influence rainfall patterns, expressed as probabilistic distributions, under both normal and extreme regimes. Wang (2009) used the Markov chain Monte Carlo (MCMC) methods to study a comparison between non-Bayesian State Space Models and Bayesian SSMs utilizing the parameter estimation strategy. Bayesian Structural Time Series (BSTS) models represent a viable alternative because they combine transparency, flexibility, and the capacity to incorporate previous information. Unlike traditional models, BSTS do not use differencing or explicit lag structures. The Bayesian framework enables robust quantification of uncertainty using posterior distributions, resulting in more reliable

predictions (Goller and Schueller 2011; Scott and Varian 2014). Gupta et al. (2011) used Bayesian structural methodology for forecasting the US real house price index to estimating the microeconomics scenario. The extreme volatile Bitcoin currency also estimated by Bayesian structural time series (BSTS) model using Spike and Slab method (Poyser 2019). The geopolitical risk on prices such as oil and gas under Russia-Ukraine war also studied using this dynamic methodology (Khurshid et al. 2024). Basher et al. (2019) studied Bayesian model averaging (BMA) model technique to predict extreme rainfall in Northeast Bangladesh. So, we incorporated Bayesian structural time series (BSTS) model for estimating the nonlinear seasonal series prediction. Also, exogenous variables allow the model to better capture the influence of external influences, potentially boosting prediction performance. In this work, BSTS model is also examined with and without exogenous factors (X), such as minimum and maximum temperature, to determine their impact on rainfall series forecasting accuracy.

All the statistical and machine learning models were evaluated using some performance indicators, such as Root Mean Square Error (RMSE), Mean Absolute Percentage Error (MAPE), Mean Absolute Error (MAE) and Mean Absolute Scaled Error (MASE). The novelty of this study lies in the comprehensive application of BSTS and BSTSX models to seasonal climatic time series i.e. rainfall, maximum temperature, and minimum temperature over a 122-year period for India, and in its rigorous comparison with both classical statistical and modern machine learning methods under a unified evaluation framework. This work contributes to the literature on Bayesian methods in climate forecasting, an area that remains underexplored despite its potential for robust uncertainty quantification. The results of this study could guide future developments in time series forecasting techniques and improve decision-making in industries that are susceptible to climate change. The Fig. 1 is showcasing the overall frame work of this research. The remaining portion of the manuscript has been divided into four sections: “2”, “3”, “4” and “5”.

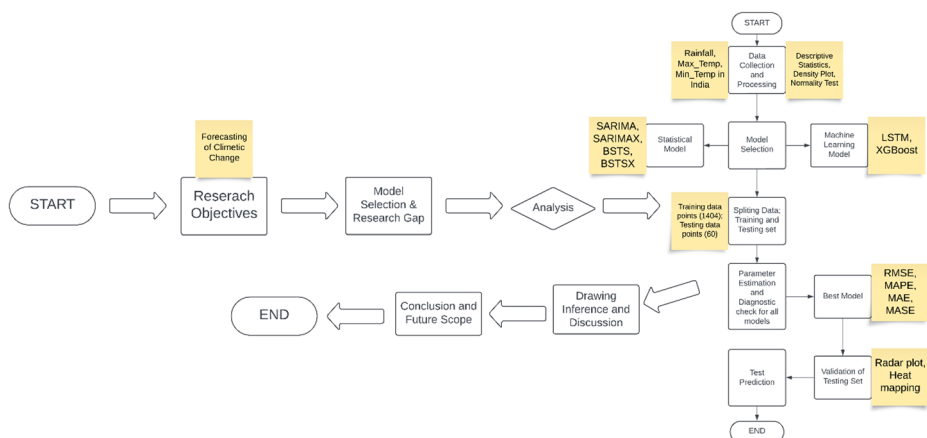


Fig. 1 The framework of the study

2 Materials and Methods

2.1 Study Information

Monthly time series data for rainfall (mm), maximum temperature (°C), and minimum temperature (°C) for India were collected from the World Bank Climate Change Knowledge Portal (<https://climateknowledgeportal.worldbank.org>; accessed July 2024) for the period January 1901 to December 2022. A total of 1464 observations were obtained for each variable. The dataset covers the entire Indian subcontinent and represents area-averaged monthly climatic conditions. The data were divided into a training set (1404 data points) and a testing set (60 data points, corresponding to January 2017 to December 2022) for model development and validation. Prior to splitting, data pre-processing was performed using the Grubbs test and boxplot visualisation, which confirmed the absence of missing values and outliers in each series (Fig. 2).

2.2 Model Description (Statistical & ML)

2.2.1 ARIMA Models

In 1970, Box and Jenkins introduced the ARIMA model as an extension of the ARMA model. The autoregressive component of ARIMA is used to predict future values by examining the linear relationship between past and present observations. The parameter “p” indicates the number of lagged observations utilized in the autoregression. ARIMA models employ differencing to transform non-stationary time series into stationary ones, effectively removing trends and seasonality. This differencing step is denoted by “d,” which signifies how many times the process is applied to achieve stationarity. To handle noise and random variations, the moving average component of ARIMA focuses on the relationship between

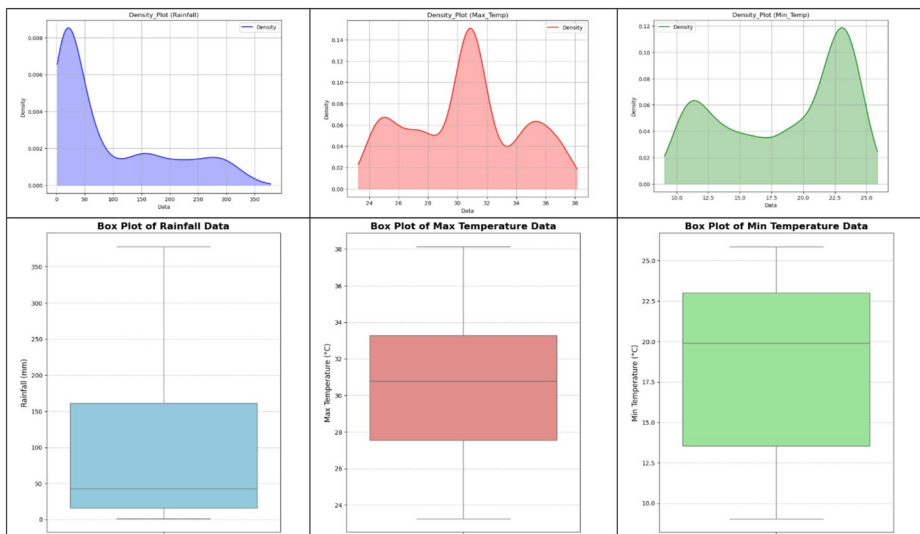


Fig. 2 Density plot and box plot for all three series

observations and residual errors from previous lags. The moving average order, denoted by “q,” specifies the number of lagged forecast errors used in the prediction. The general form of the forecasted value at time t generated by the ARMA (p, q) process can be expressed as:

$$y_t = c + \alpha_1 y_{t-1} + \dots + \alpha_p y_{t-p} + \epsilon_t - \beta_1 \epsilon_{t-1} - \dots - \beta_q \epsilon_{t-q} \quad (1)$$

where α_i ($i=1,2,\dots,p$) and β_j ($j=1,2,\dots,q$) are the model parameters.

2.2.2 Seasonal ARIMA(SARIMA)

SARIMA is the extension of ARIMA model when the time series exhibits a seasonal variation. The SARIMA model is generally expressed as SARIMA(p, d, q)(P, D, Q)_s. Here P denotes the number of seasonal autoregressive terms, D denotes the number seasonal differencing and Q denotes the number of Seasonal moving average. The subscripted letter ‘s’ denotes the length of the seasonal cycle. For example, $s=7$ or 4 or 12 according to the time series is in an hourly or in a quarterly or in a monthly data respectively. The general form of SARIMA(p, d, q)(P, D, Q)_s is given by (Permanasari et al. 2013).

$$\phi_p(B) \phi_P(B^S) Z_t = \delta + \theta_q(B) \theta_Q(B^S) a_t \quad (2)$$

Z_t is the time series data at time t after it has been differenced to remove trends and seasonality, which is given by

$$Z_t = (1 - B^S)^D (1 - B)^d y_t \quad (3)$$

‘d’ is the degree of non-seasonal differencing and D is the degree of seasonal differencing. B is the backshift or lag operator.

If d times differencing is needed for the data to become stationary, then the ARIMA model is represented as ARIMA (p, d, q), where p and q refer to the order of the autoregression and moving average, respectively. Model parameters are estimated by the method of maximum likelihood.

2.2.3 Bayesian Structural Time Series Model

The Bayesian Structural Time Series (BSTS) model is considered more transparent than the traditional Autoregressive Integrated Moving Average (ARIMA) framework because it explicitly decomposes a time series into interpretable components such as trend, seasonality, and regression effects. Unlike ARIMA models, whose representation depends on differencing, autoregressive lags, and moving-average terms, BSTS models allow direct visual inspection of the underlying structural components.

Another important advantage of BSTS models is their principled treatment of uncertainty. The Bayesian framework enables quantification of posterior uncertainty associated with individual components, incorporation of prior information, and regularization of regression coefficients through shrinkage priors. Moreover, any ARIMA model can be represented in an equivalent structural state-space form.

A general BSTS model may be expressed as:

$$y_t = \mu_t + x_t\beta + S_t + e_t, e_t \sim N(0, \sigma_e^2) \quad (4)$$

Where:

y_t : observed time series at time t

μ_t : is the local level term

x_t : denotes a set of regressors

β vector of regression coefficients

S_t : represents seasonality

The local level evolves according to:

$$\mu_{t+1} = \mu_t + v_t, v_t \sim N(0, \sigma_v^2) \quad (5)$$

The regression coefficients, trend, and seasonal effects are estimated simultaneously, thereby reducing the possibility of spurious regression relationships. Owing to the Bayesian formulation, prior distributions may be imposed on the regression coefficients to induce sparsity or incorporate expert knowledge.

The following specifications were adopted in the present study:

- Total Markov Chain Monte Carlo (MCMC) draws: 500
- Burn-in period: First 100 iterations discarded
- Prior inclusion probability: 0.5 (Spike-and-Slab prior for BSTS regression component)
- Holdout period: Last 60 observations reserved for testing
- Final forecasts obtained by averaging posterior samples after burn-in removal

Prior Specification Let γ denote the collection of model parameters and let $\varphi = \varphi_1, \dots, \varphi_n$ represent the latent state sequence. Suppose the prior distribution of the parameters is denoted by $p(\gamma)$, and the initial state distribution is denoted as $p(\varphi_0|\gamma)$. Using Bayes' theorem, posterior inference is based on $p(\varphi, \gamma|d)$. Since the posterior distribution is analytically intractable, Markov Chain Monte Carlo (MCMC) methods are employed for sampling. For variable selection in the regression component, the Spike-and-Slab prior is adopted. Let τ denote the vector of regression coefficients and define the binary inclusion indicator: $\theta_t = 1, \tau_t \neq 0$ and $\theta_t = 0$, otherwise. Also, if τ_n expresses the nonzero scalars for the vector τ . Further, if the sequence $q = q_1, \dots, q_n$ and $\sum_q^{-1}$ are the rows and columns with nonzero entries for q , we can write the spike and slab prior as follows (Xie 2021):

$$p\left(q, \tau, \frac{1}{\sigma_\varepsilon^2}\right) = p(q) p(\sigma_\varepsilon^2|q) p(\tau_m|q, \sigma_\varepsilon^2) \quad (6)$$

Assuming a Gaussian slab prior, we obtain:

$$\tau_q | \sigma_\varepsilon^2 \sim N(b_q, \sigma_\varepsilon^2 (\Sigma_q^{-1})^{-1}) \quad (7)$$

Posterior Specification For the full likelihood function, if the law of total probability is $p(A) = \sum_n P(A|B_n) P(B_n)$, then we have the function:

$$p\left(\delta, \tau, \sum q \pi\right)=p\left(\delta, \tau, \sum q \pi\right) p\left(\tau \mid \mu\right) p\left(\sum q \mid \tau\right) p(\pi) \quad (8)$$

Here, δ expresses the set of values to multiple time series that is components of time series with trend, seasonality and others. Also, $\sum q$ is a correlated error. The Bayesian framework standardizes the observations, incorporates prior beliefs, and generates posterior distributions through MCMC sampling, thereby enabling probabilistic forecasting and uncertainty quantification.

2.2.4 Long Short-term Memory Models

Long Short-Term Memory (LSTM) networks, introduced by Sepp Hochreiter and Jürgen Schmidhuber in 1997, are a specialized type of Recurrent Neural Network (RNN). They offer greater flexibility than traditional RNNs and are specifically designed to address the issue of vanishing gradients, a common problem in standard RNNs. LSTMs are particularly effective for handling sequential data, such as time-series data, because they can learn long-term dependencies. This capability is made possible by their core components called memory cells. Each memory cell contains three types of gates: the input gate, forget gate, and output gate. These gates control the flow of information based on its relevance, enabling the network to determine what to retain, what to discard, and what to prioritize. The calculation process of LSTM is given as follows.

Candidate memory cell $\tilde{c}$ is calculated, and is given by

$$\tilde{c}=\tanh \left(W_c\left[h_{k-1}, X_k\right]+B_c\right) \quad (9)$$

W_c is the weight matrix, B_c is the bias.

Input gate is calculated by the following formula

$$i_k=\sigma\left(W_i\left[h_{k-1}, X_k\right]+B_i\right) \quad (10)$$

The input gates use sigmoid activation function which gives values between 0 and 1, so that it helps in making decision of the lstm unit to disregard or select information in the cell state accordingly.

Forget gate is calculated by

$$f_k=\sigma\left(W_f\left[h_{k-1}, X_k\right]+B_f\right) \quad (11)$$

The forget gate decides which part of the previous cell state should be retained or discarded.

Output gate controls the flow of information from the memory cell to the output of the model

$$o_k=\sigma\left(W_o\left[h_{k-1}, X_k\right]+B_o\right)$$

Cell state serves as a memory unit that enables the LSTM to gather and keep track of important data across long sequences.

$$c_k=\left(i_k * \tilde{c}+f_k * c_{k-1}\right) \quad (12)$$

New state - $h_k = o_k * \tanh(c_k)$.

X_k – Input gate

h_k – output vector

2.2.5 XGBoost Model

XGBoost (Extreme Gradient Boosting) is a scalable and efficient tree-boosting algorithm. It was formally introduced by Chen and Guestrin (2016), and has since become a widely used tool for regression and forecasting tasks. When applied to time series analysis, XGBoost treats the problem as a regression task where past data points are used as features to forecast future values. Its iterative learning approach, combined with its ability to handle missing data, helps in simplifying the model while maintaining accuracy. The regularization aspect of XGBoost is particularly beneficial in time series forecasting, as it helps avoid overfitting to noise or repetitive seasonal patterns.

Moreover, XGBoost's ability to process large and complex datasets efficiently makes it an excellent choice for time series forecasting tasks. Its support for parallel and distributed computation enables fast processing of extensive time series data, making it suitable for large-scale forecasting applications.

XGBoost sums the results of K trees as the final predicted value, as follows:

$$\hat{y}_t = \phi(X_i) = \sum_{k=1}^K f_k(X_i), \quad f_k \in F, \quad (13)$$

Where $\hat{y}_t$ represents the predicted value, X_i represents the i -th sample and F represents the space of the regression tree. F is given by

$$F = \{f(X) = \omega_{q(X)}\} (q: \mathbb{R}^m T \rightarrow \omega \in \mathbb{R}^T) \quad (14)$$

Here T denotes the number of leaf nodes and q represents the structure of each tree. Again the objective function which is to be minimized is given by

$$\mathcal{L}(\phi) = \sum_i l(\hat{y}_i, y_i) + \sum_k \Psi(f_k) \quad (15)$$

$\sum_i l(\hat{y}_i, y_i)$ is the mean square error and $\sum_k \Psi(f_k)$ is a regular term that controls the complexity of the tree and prevents overfitting.

$$\Psi(f) = \gamma(T) + \frac{1}{2} \lambda \|\omega\|^2 \quad (16)$$

The model is trained in an additive manner based on the above objective function. So finally the objective function becomes after t -th round.

$$\mathcal{L}^t = \sum_{i=1}^n l(\hat{y}_i^{(t-1)} + f_t(X_i), y_i) + \Psi(f_t) \quad (17)$$

After a series of improvements the final equation which is used to score a leaf node during splitting is given by

$$Gain = \frac{1}{2} \left[\frac{G_L^2}{H_L + \lambda} + \frac{G_R^2}{H_R + \lambda} - \frac{(G_L + G_R)^2}{H_L + H_R + \lambda} \right] - \gamma \quad (18)$$

Where $\frac{G_L^2}{H_L + \lambda}$ is the score on the left leaf, $\frac{G_R^2}{H_R + \lambda}$ is the score on the right leaf and $\frac{(G_L + G_R)^2}{H_L + H_R + \lambda}$ is the score on the original leaf.

2.3 Software

All analyses were conducted in Python. Software and packages version numbers are provided in supplementary material. Packages used include pmdarima (SARIMA/SARIMAX), statsmodels (diagnostic testing), tensorflow and keras (LSTM), xgboost (XGBoost), and tensorflow-probability (BSTS).

3 Results

3.1 Data Description

As standard protocol goes we start with the descriptive of the datasets being used for the present investigation. Table 1 presents the important statistics of the datasets, which revealed that the rainfall series having highest variation (CV % :105.927) and the maximum temperature having lowest variation of 12.412 (CV%). The values of the kurtosis revealed the datasets to be platykurtic in nature with rainfall and maximum temperature series being positively skewed and minimum temperature series exhibiting negative skewness. Further, density plots (Fig. 2) of the datasets along with the Shapiro-Wilk test statistic values distinctively reveals deviation from normality. Under pre-processing of the data series, no outlier were found, confirmed by the Grubbs test and boxplot (Fig. 2). Overall, from the descriptive statistics and density plot we conclude that the datasets under consideration are non-normal in nature. These characteristics guided the selection of appropriate forecasting models.

3.2 Model Development (Statistical & ML)

The analysis begins with the rainfall series as the primary variable of interest. Initially SARIMA and SARIMAX models were incorporated which falls under category of classical statistical model. The appropriate models were identified along with their parameter

Table 1 Descriptive statistics

	Descriptive Statistics		
	Rainfall	Max_Temp	Min_Temp
Min	1.25	23.26	9.04
Max	378.01	38.13	25.86
Mean	93.6317	30.538	18.5147
SD	99.1816	3.7904	4.9951
CV%	105.9274	12.412	26.9791
Skew	1.0131	0.00386	-0.4008
Kurtosis	-0.36007	-0.85744	-1.3698
Shapiro–Wilk test	0.80964	0.95974	0.8865

estimation was reported in Table 2. To develop the SARIMA and SARIMAX model, first `auto_arima` library was used from `pmdarima` package to find best parameter (p, d, q)(P, D, Q) combination of the models. Then `ARIMA` and `ARIMAResults` library from `statsmodels` were used to build the models with specific parameters (p, d, q)(P, D, Q). Additionally, for SARIMAX model, `exog` used to incorporate the exogenous variable for building the model.

The SARIMA (0,0,2) (1,0,1) model with intercept was found to be appropriate for the rainfall series. The parameter estimates along with standard error (S.E) indicates the statistical significance of the estimates obtained, also the Ljung-Box Test (LB) statistic for lags 5 and 10 clearly show the absence of autocorrelation of the residuals at these lags. Thus we can safely interpret that the SARIMA model is a good fit for the seasonal rainfall series. Next, in order to improve the modelling and forecasting efficiency we have made use of SARIMAX model. In this model we have used maximum and minimum temperature series as exogenous variables influencing the rainfall series. The SARIMAX (0,0,2) (1,0,1) with intercept model was found to be best suited. The regressor maximum temperature seems to influence the rainfall series positively with an estimated coefficient value to be 0.4529 and the minimum temperature is found to have negative impact on rainfall to the tune of -0.387. Both, the regressors coefficients were found to be statistically significant as indicated from the values of corresponding S.E. It is to be noted that the value of Sigma, which explains the total error variance has decreased in SARIMAX model (0.165) as compared to SARIMA model (0.1708) which is indicative of that fact that SARIMAX model is efficient in modelling the rainfall data as compared to SARIMA model. As in case of SARIMA model residuals of SARIMAX model also indicates no presence of autocorrelation in lags 5 and 10 respectively. Also the diagnostic plot was satisfied that the residuals obtained by both the models followed normal distribution (Fig. 3). Further, we attempt to model the maximum and minimum temperature series individually using SARIMA model. For maximum temperature series SARIMA (1,1,2) (0,0,2) with intercept was found to be appropriate. The parameter estimates are reported in Table 3 along with corresponding S.E. For adequacy check of the fitted model, LB test was conducted which revealed that the residuals were free from any autocorrelation at lags 5 and 10. In case of minimum temperature series SARIMA

Table 2 Parameter information of rainfall

	Parameter Estimation			Ljung-Box Test		
	Coefficient	Estimate	SE	Lags point	Statistic	P value
SARIMA(0,0,2)(1,0,1) intercept	Intercept	3.8272	0.28	5	5.730756	0.220178
	ma1	0.0329	0.007	10	16.48537	0.0865
	ma2	-0.056	0.012			
	ar_s1	0.9999	7.83E-05			
	ma_s1	-0.9707	0.01			
	sigma	0.1708	0.004			
SARIMAX(0,0,2)(1,0,1) intercept	Intercept	7.2931	0.361	5	17.751	0.3275
	Max_temp	0.4529	0.015	10	21.0675	0.2062
	Min_temp	-0.387	0.019			
	ma1	0.1244	0.021			
	ma2	-0.1142	0.022			
	ar_s1	0.9138	0.024			
	ma_s1	-0.7904	0.034			
	sigma	0.165	0.005			

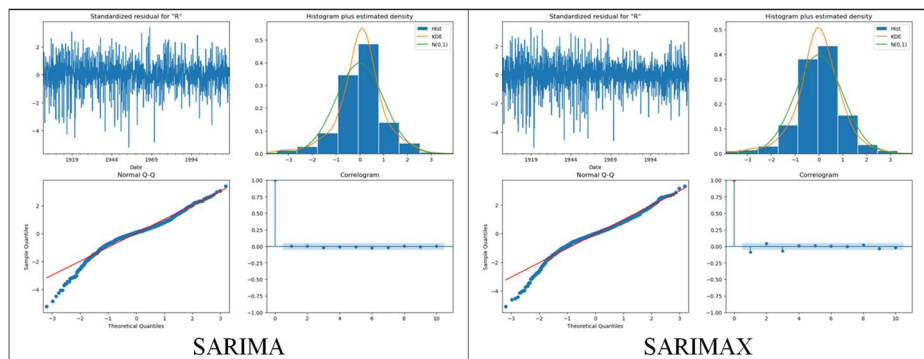


Fig. 3 Diagnostic plot of rainfall series

Table 3 Parameter information of Max_Temp

	Parameter Estimation			Ljung-Box Test		
	Coefficient	Estimate	SE	Lags point	Statistic	<i>P</i> value
SARIMA(1,1,2)(0,0,2)	ar1	-0.3092	0.169	5	7.563	0.1453
	ma1	0.6225	0.164	10	11.239	0.1265
	ma2	0.2347	0.046			
	ma_s1	0.6329	0.023			
	ma_s2	0.4721	0.024			
	sigma	2.1284	0.073			

Table 4 Parameter information of Min_Temp

	Parameter Estimation			Ljung-Box Test		
	Coefficient	Estimate	SE	Lags point	Statistic	<i>P</i> value
SARIMA(1,0,0)(1,0,1) intercept	Intercept	18.6224	1.065	5	8.261	0.119
	ar1	0.3381	0.023	10	14.651	0.0861
	ar_s1	0.9998	7.73E-05			
	ma_s1	-0.9067	0.013			
	sigma	0.3204	0.011			

(1,0,0) (1,0,1) model was identified to best suited. The details of the model estimates along with S.E. are reported in Table 4. As done earlier, the residuals of the model were tested for presence of autocorrelation and LB test confirmed the absence of any autocorrelation for lags 5 and 10 respectively. The residuals also followed normal distribution depicted by the diagnostic plot in both the series (Fig. 4).

After analysing the datasets using statistical models (SARIMA and SARIMAX) we turn our attention towards machine learning and deep learning models. For the present study we have selected LSTM and XGBoost models based on extensive literature survey. All the 3 series were modelled and forecasted using these 2 selected models. Table 5 contains the list of hyperparameters that were found optimum for the datasets under consideration for both the models. The number of lag in case of LSTM model was selected using ACF criteria and other hyperparameters were selected based on trial-error technique. The feature lags selected for all the series was 6 with 350 neurons, activation function relu, dense layer

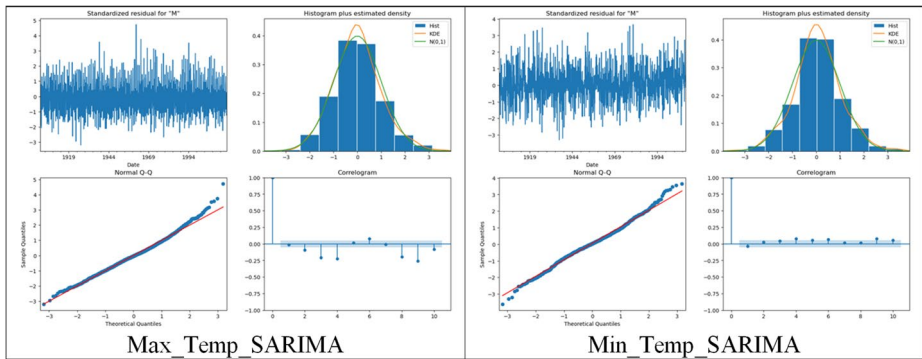


Fig. 4 Diagnostic plot of Max and Min Temp

Table 5 Parameter information of LSTM and XGBoost model

	Feature Lag	Neuron	Dense_Layer		Total Param	Epochs
			Layer_1	Layer_2		
Rainfall	6	350	2	1	4,93,505	35
Max_Temp						
Min_Temp						
XGBoost	n_estimator					
XGBRegressor	100					

Fig. 5 XGBoost lag feature

```
# Function to create lag features
def create_lag_features(df, lag):
    df_lag = pd.DataFrame(df[['Rainfall']])
    for i in range(1, lag + 1):
        df_lag[f'lag_{i}'] = df_lag['Rainfall'].shift(i)
    df_lag.dropna(inplace=True)
    return df_lag
```

combination (2, 1). The model compiled with adam optimizer and mse loss function, total param formed 493,505. The model then fitted with trained scaler set (scaled with MinMaxScaler()) with 35 epochs. For XGBoost model development, XGBRegressor library from xgboost package was used, with same feature lag 6 and n_estimator 100 for all the series. To create the lag feature for XGBoost model, the simple python code implies depicted in Fig. 5.

Now, we finally focus on Bayesian models i.e., BSTS and BSTSX. The details of the model setup are given in Table 6. The selection of prior distribution was decided based on review of literature. The MCMC draws and burn-in period was obtained based on the convergence of the algorithm and stability of the estimates obtained.

3.3 Model Validation

After identifying models for the datasets we have compared their performances in training and testing sets based on popular error measures namely RMSE, MAE, MAPE and MASE. Table 7 presents the values pertaining to each error measure for each model.

Table 6 Details of Bayesian model

Model	Coefficients	Estimate
BSTSX	Maximum Temperature	-27.567
	Minimum Temperature	18.255
Prior	Spike and Slab, MCMC draws: 500	

Table 7 Model evaluation of rainfall series

Model	Training Set				Testing Set			
	Goodness of fit				Goodness of fit			
	RMSE	MAE	MAPE	MASE	RMSE	MAE	MAPE	MASE
SARIMA(0,0,2)(1,0,1) intercept	25.238	16.084	34.757	0.325	28.936	18.655	30.173	0.377
SARIMAX(0,0,2)(1,0,1) intercept	28.001	17.343	33.774	0.350	29.151	17.870	30.368	0.361
LSTM	69.905	46.847	71.088	0.946	76.382	51.107	64.639	1.032
XGBoost	6.863	4.862	19.956	0.098	28.571	21.018	39.972	0.425
BSTSX	19.763	13.953	42.893	0.282	25.506	17.268	25.976	0.320
BSTS	21.826	14.832	39.246	0.300	25.804	17.628	25.867	0.319

Rainfall Series To begin with we start with the results obtained on training set. Here we notice LSTM model to perform poorly among all the models in all four error measures (RMSE: 69.905; MAE: 46.847; MAPE: 71.088; and MASE: 0.946) followed by SARIMAX model (RMSE: 28.001; MAE: 17.343; MAPE: 33.774; and MASE: 0.350). SARIMA model (RMSE: 25.238; MAE: 16.084; MAPE: 34.757; and MASE: 0.325) performed slightly better as compared to SARIMAX model. The performance of BSTS and BSTSX models were close with BSTSX model (RMSE: 19.763; MAE: 13.953; MAPE: 42.893; and MASE: 0.282) exhibiting superior performance as compared to BSTS model (RMSE: 21.826; MAE: 14.832; MAPE: 39.246; and MASE: 0.300) in all four error measures. XGBoost stood out to be the best in terms of training set performance with lowest values for the error measures (RMSE: 6.863; MAE: 4.862; MAPE: 19.956; and MASE: 0.098). Now we turn our attention towards testing set performance which holds immense importance as it will throw light into the generalisation ability of the models considered. As in the training set LSTM, SARIMA and SARIMAX model performed in similar lines for testing sets also with RMSE values of 76.382, 28.936, 29.151; MAE values of 51.107, 18.655, 17.870; MAPE values of 64.639, 30.173, 30.368 and MASE values to be 1.032, 0.377, 0.361 respectively for the three models. But, when we look into the performance of XGBoost (RMSE: 28.571; MAE: 21.018; MAPE: 39.972; and MASE: 0.425) it is evident that the model had over-fitted in the training set, hence failed to keep up its performance in the testing set. Here, also BSTS and BSTSX models performed closely in all four error measures, with BSTSX model (RMSE: 25.506; MAE: 17.268; MAPE: 25.976; and MASE: 0.320) surpassing the performance of BSTS model (RMSE: 25.804; MAE: 17.628; MAPE: 25.867; and MASE: 0.319). thus, we document the superior performance of Bayesian models over other class of competing models.

Maximum Temperature For this dataset as we are modelling it singularly we have 4 models (SARIMA, LSTM, XGBoost and BSTS) to be compared. The values thus obtained on error measures for training and testing sets are reported in Table 8. If we look into the training set performances of the 4 models, we could identify that the as earlier LSTM model performed

Table 8 Model evaluation of Max_temp series

Model	Training_Set				Testing_Set			
	Goodness of fit				Goodness of fit			
	RMSE	MAE	MAPE	MASE	RMSE	MAE	MAPE	MASE
SARIMA(1,1,2)(0,0,2)	1.597	1.148	3.811	0.520	7.960	7.033	21.703	3.184
LSTM	2.575	2.140	6.580	0.969	2.746	2.324	7.050	1.052
XGBoost	0.193	0.145	0.483	0.066	0.741	0.595	1.911	0.269
BSTS	0.503	0.388	1.281	0.176	0.593	0.485	1.594	0.220

Table 9 Model evaluation of Min_temp series

Model	Training_Set				Testing_Set			
	Goodness of fit				Goodness of fit			
	RMSE	MAE	MAPE	MASE	RMSE	MAE	MAPE	MASE
SARIMA(1,0,0)(1,0,1) intercept	0.675	0.460	2.823	0.197	0.516	0.394	2.230	0.169
LSTM	0.967	0.899	5.027	0.385	1.026	0.967	5.263	0.415
XGBoost	0.195	0.146	0.890	0.063	0.709	0.522	2.776	0.224
BSTS	0.476	0.369	2.246	0.158	0.614	0.520	2.935	0.226

poorly with highest values of error measures (RMSE: 2.575; MAE: 2.140; MAPE: 6.580; and MASE: 0.969). This was followed by SARIMA model (RMSE: 1.597; MAE: 1.148; MAPE: 3.811; and MASE: 0.520). BSTS model had RMSE of 0.503; MAE of 0.388; MAPE of 1.281 and MASE of 0.176. finally, XGBoost was found to best for training set with lowest values of error measures (RMSE: 0.193; MAE: 0.145; MAPE: 0.483; and MASE: 0.066). Now turning our attention towards testing set, we document similar results for LSTM and SARIMA model, but here LSTM (RMSE: 2.746; MAE: 2.324; MAPE: 7.050; and MASE: 1.052) outperformed the SARIMA model (RMSE: 7.960; MAE: 7.033; MAPE: 21.703; and MASE: 3.184). and again the BSTS model (RMSE: 0.593; MAE: 0.485; MAPE: 1.594; and MASE: 0.220) stood out to be the best model surpassing the XGBoost model (RMSE: 0.741; MAE: 0.595; MAPE: 1.911; and MASE: 0.269).

Minimum Temperature As for the maximum temperature series, we have also modelled this series individually. The results of the training set (Table 9) are sink with that of maximum temperature with LSTM model performing worst (RMSE: 0.967; MAE: 0.899; MAPE: 5.027; and MASE: 0.385). This was followed by SARIMA model (RMSE: 0.675; MAE: 0.460; MAPE: 2.823; and MASE: 0.197) and then BSTS model had RMSE of 0.476; MAE of 0.369; MAPE of 2.246 and MASE of 0.158. Finally, XGBoost was found to best for training set with lowest values of error measures (RMSE: 0.195; MAE: 0.146; MAPE: 0.890; and MASE: 0.063). But, for testing set, LSTM (RMSE: 1.026; MAE: 0.967; MAPE: 5.263; and MASE: 0.415) again performed poorly followed by XGBoost (RMSE: 0.709; MAE: 0.522; MAPE: 2.776; and MASE: 0.224) but SARIMA achieved marginally lower value (RMSE: 0.516; MAE: 0.394; MAPE: 2.230; and MASE: 0.169) compared to BSTS (RMSE: 0.614; MAE: 0.520; MAPE: 2.935; and MASE: 0.226), making SARIMA the best-performing model for this particular series. Hence, for this series SARIMA model was found to be superior to other competing models. We incorporated DM test to check the significance of both the models (SARIMA and BSTS) and found that both models are *at par*.

For ease of understanding we have added plots for enhanced visibility of the results obtained. The predicted values along with original values obtained from each models are depicted through Fig. 6. From this figure one can clearly identify the appropriateness of BSTS and BSTSX model for the rainfall series. Similar figures have also been reported for maximum temperature (Fig. 7) and minimum temperature (Fig. 8). All these figures together help us to understand the capabilities of different models for predicting these datasets. For further clarity of the reported results we have radar plot (Fig. 9) and heat map (Fig. 10) of the testing datasets. Both these figures aid us to establish the superiority of the Bayesian models (BSTS and BSTSX) for the datasets under consideration. The different states obtained from BSTS and BSTSX models are reported in Fig. 11. The trend component captures the long-term direction of each series. The seasonal component reflects the monthly periodic pattern with a period of 12 months and the regression component in BSTSX quantifies the contribution of maximum and minimum temperature as exogenous predictors to the rainfall level. These components collectively demonstrate how the BSTS framework decomposes the observed series into interpretable sub-structures.

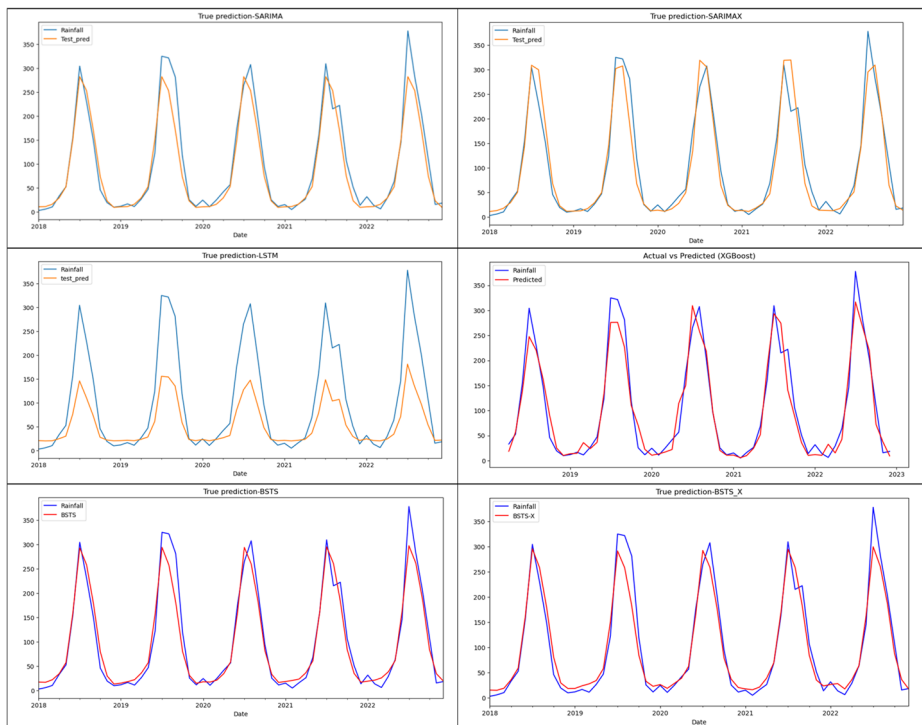


Fig. 6 Test_set prediction for rainfall

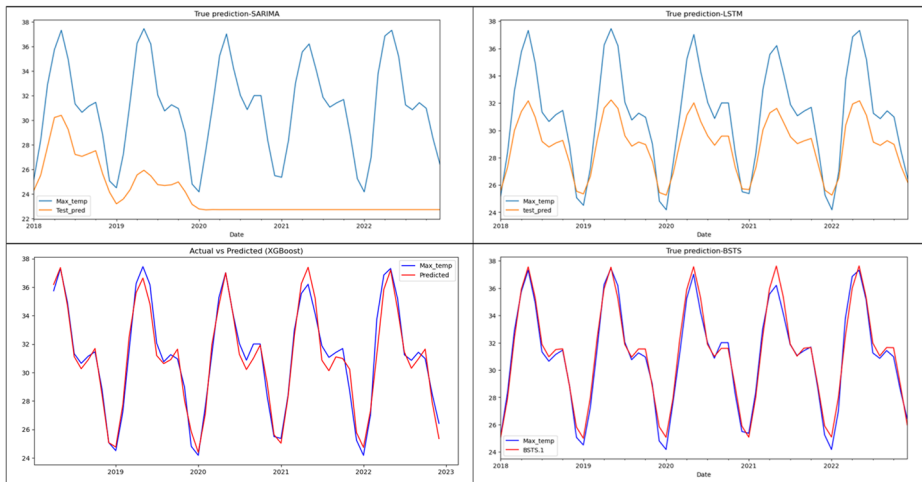


Fig. 7 Max_Temp test set prediction

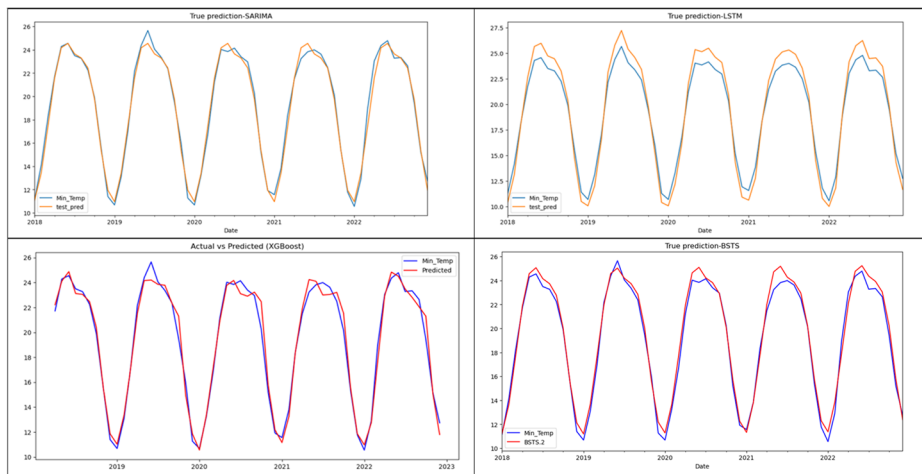


Fig. 8 Min_Temp test set prediction

4 Discussion

The results of this study provide several important insights into the relative performance of Bayesian, statistical, and machine learning models for seasonal climate forecasting. The poor performance of the LSTM model across all three variables, particularly on the training set for rainfall (RMSE: 69.905; MAPE: 71.088%) is noteworthy given the widespread enthusiasm for deep learning in climate forecasting. This result is consistent with findings by Barrera-Animas et al. (2022), who observed that LSTM performance is highly sensitive to data volume and hyperparameter tuning, particularly for strongly seasonal series. Pre-processing using seasonal decomposition or deseasonalisation explored by Chatatidis et

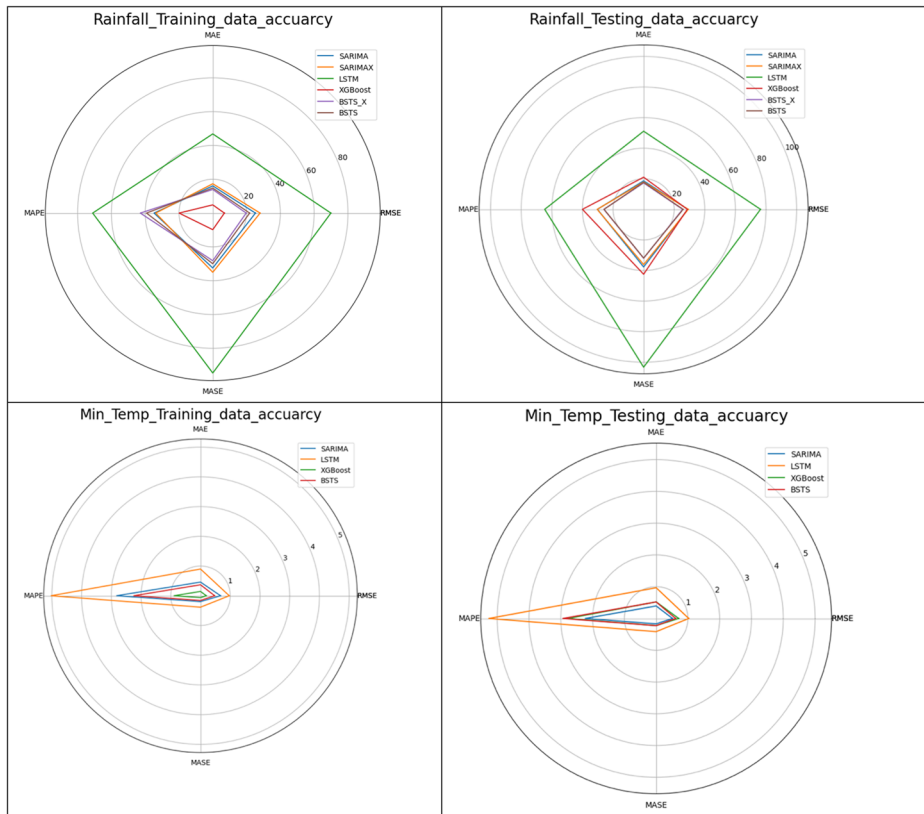


Fig. 9 Radar plot for model accuracy visualization

al. (2023) may improve LSTM performance and is identified as a future research direction. XGBoost achieved excellent training set performance across all three series (e.g., RMSE: 6.863 for rainfall; RMSE: 0.193 for maximum temperature) but exhibited pronounced overfitting on the testing set. For rainfall, the testing-set RMSE of XGBoost (28.571) was more than four times greater than its training-set RMSE (6.863), a discrepancy consistent with the well-documented tendency of gradient boosting methods to overfit when seasonal lag features are constructed from short-memory time series (Mishra et al. 2024; Dong et al. 2023). The inclusion of regularisation strategies or expanding the feature engineering pipeline may help mitigate this issue. The superior generalisation of BSTS and BSTSX on the rainfall and maximum temperature testing sets aligns with the theoretical advantages of Bayesian structural modelling. As Scott and Varian (2014) examined the explicit decomposition of trend, seasonality, and exogenous effects within a probabilistic framework reduces the risk of overfitting while preserving interpretability. The BSTSX model incorporates maximum and minimum temperature as exogenous predictors, achieved slightly lower testing set RMSE (25.506) than BSTS (25.804) for rainfall, confirmed that temperature variables carry predictive information for monsoon dynamics. This finding is in agreement with Shah et al. (2024) and Mulla et al. (2024), who highlighted the value of temperature as an exogenous covariate in seasonal rainfall models. The negative coefficient of minimum temperature

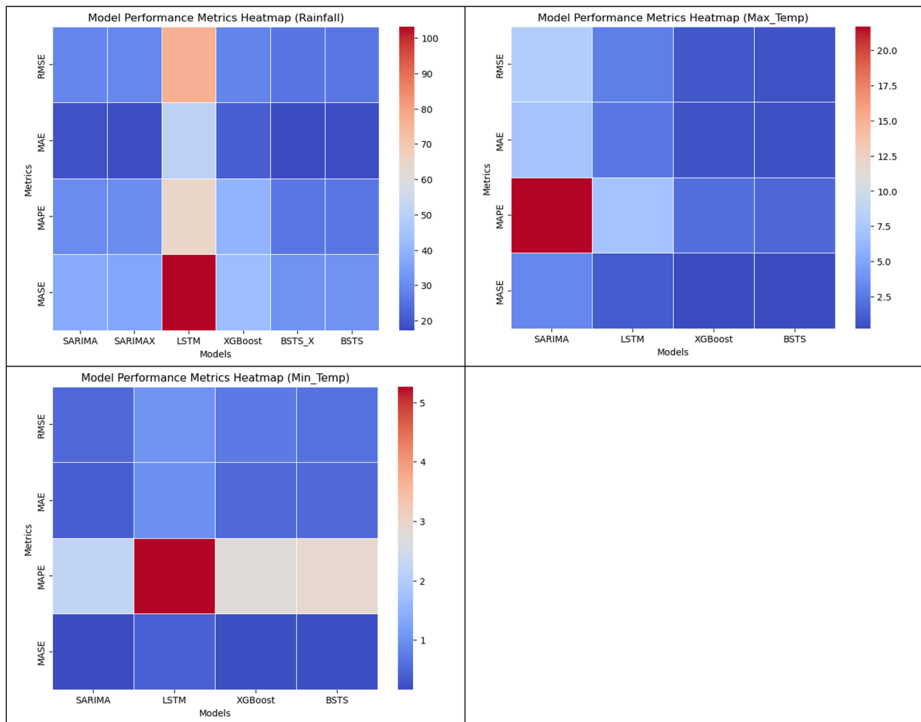
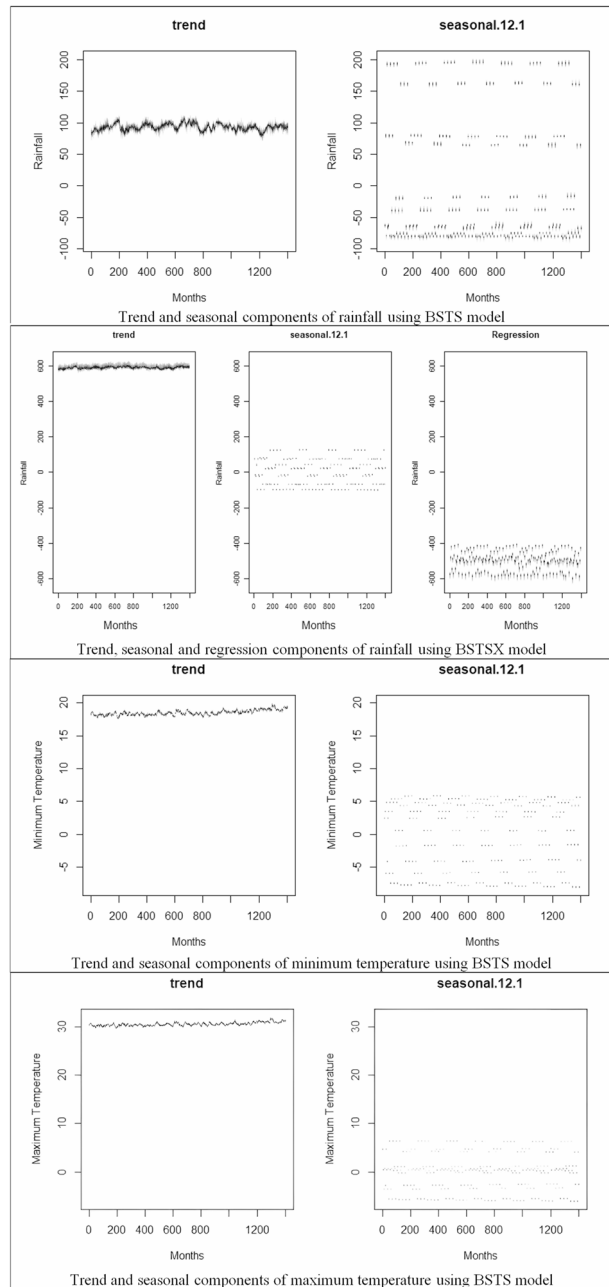


Fig. 10 Heatmap of testing set visualization

(-27.567) in the BSTSX model may reflect the inverse relationship between cool season temperatures and warm season rainfall in the Indian subcontinent, a relationship that warrants further investigation.

The exception for minimum temperature, where SARIMA(1,0,0)(1,0,1) outperformed BSTS on the testing set (RMSE: 0.516 vs. 0.614; MAPE: 2.230% vs. 2.935%). The minimum temperature series exhibits relatively low variability ($CV\% = 26.98\%$) and strong autoregressive structure (AR1 coefficient: 0.338; AR_S1 coefficient ≈ 1.00), suggested that a parsimonious linear model may be sufficient to capture its dynamics. This finding is consistent with the view that BSTS models provide the greatest benefit when non-linearity, structural breaks, or informative prior knowledge can be exploited for relatively smooth, low-variance series, the classical models may remain competitive (Harrison and Stevens 1971; Das and Chanda 2020). The difference between the two models ($\Delta RMSE = 0.098$; $\Delta MAPE = 0.705\%$) is modest, and the BSTS model retains the advantage of uncertainty quantification through posterior distributions. Future work employing informative priors based on climate domain knowledge may further improve BSTS performance for this series.

The ability of BSTS and BSTSX to provide probabilistic forecasts rather than the point estimates is particularly valuable. Probabilistic rainfall forecasts can directly inform reservoir operations, irrigation scheduling, and drought early-warning systems. Uncertainty quantification is especially critical in the Indian context, where the inter-annual variability of monsoon rainfall ($CV\% = 105.93\%$) creates substantial risk for agricultural and hydrological planning.

Fig. 11 Different states obtained from BSTS and BSTSX models

The BSTSX model, by explicitly quantifying the contribution of temperature to rainfall variability, aligns with the growing recognition that climate change-driven shifts in temperature regimes will alter monsoon dynamics in the Indian subcontinent (Rawat et al. 2022; Wani et al. 2024). Adoption of BSTS based forecasting in operational hydrometeorological applications could support more adaptive and risk-aware climate resource management decisions.

5 Conclusion

This study modelled and forecasted the seasonal climatic variables such as rainfall, maximum temperature, and minimum temperature for India using a century-long monthly dataset. The primary objective was to efficiently capture the seasonal pattern of the rainfall series using temperature series as exogenous variables. Four classes of models were evaluated i.e. classical statistical (SARIMA, SARIMAX), Bayesian structural (BSTS, BSTSX), machine learning (XGBoost), and deep learning (LSTM). The LSTM model consistently underperformed, likely owing to its limited ability to capture strong seasonal patterns in the absence of explicit pre-processing. XGBoost exhibited pronounced overfitting, with testing-set errors substantially exceeding training-set performance. SARIMAX established its superiority over SARIMA by effectively incorporating exogenous temperature variables. Bayesian models (BSTS and BSTSX) demonstrated the best overall generalisation for rainfall (testing-set MAPE: approximately 26%) and maximum temperature (testing-set MAPE: approximately 1.59%). It should be noted that for minimum temperature, SARIMA achieved marginally lower testing set errors than BSTS (RMSE: 0.516 vs. 0.614; MAPE: 2.230% vs. 2.935%), indicated that for low variance, autoregressive series, classical linear models remain competitive. However, the BSTS framework retains the unique advantage of probabilistic uncertainty quantification, which is of particular importance for applications in water resource management and agricultural planning under climate change. Future work will investigate the performance of LSTM applied to deseasonalised series and the incorporation of informative priors in the BSTS framework to further improve forecast accuracy.

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Declarations

Conflict of interest The author(s) declare no competing interests.

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